

NanoGT: A Computational Framework for Matching Monogenic Rare Diseases to Gene Therapy Precedents

Lili Yassin — Imperial College London

June 2026

Table of Contents

1	Introduction	2
1.1	The Rare Disease Treatment Gap	2
1.2	Gene Therapy as a Potential Platform Solution	3
1.3	The Precedent Identification Problem	3
1.4	The NanoGT Framework	4
1.5	Paper Structure	5
2	Methods	6
2.1	1. System Overview	6
2.2	2. Disease and Gene Characterisation	6
2.3	3. Gene Therapy Surrogate Database	8
2.4	4. Twelve-Dimension Scoring Engine	8
2.5	5. Score Normalisation and Confidence Classification	13
2.6	6. Output and Report Generation	13
3	Results	14
3.1	1. Dataset Overview	14
3.2	2. Retrospective Validation	14
3.3	3. Prospective Discovery Results	16
3.4	4. Score Distribution and Summary	20
4	Figures	21
5	Discussion	23
5.1	1. Summary of Principal Findings	23
5.2	2. Validation and Framework Accuracy	24
5.3	3. Generalisability Across Disease Classes	25
5.4	4. Limitations	25
5.5	5. Clinical Translation Implications	28
5.6	6. Future Directions	29

6	References	29
6.1	Gene Therapy Reviews	30
6.2	Approved Gene Therapy Programs — Primary Clinical References	30
6.3	AAV Biology, Packaging, and Engineering	31
6.4	Rare Disease Epidemiology and Databases	32
6.5	Immunogenicity and Immune Responses to AAV	33
6.6	Lysosomal Storage Diseases and Gene Therapy	33
6.7	Haematopoietic Stem Cell Gene Therapy	34
6.8	Protein Databases and Bioinformatics Tools	34
6.9	Regulatory Guidance Documents	35

1 Introduction

1.1 The Rare Disease Treatment Gap

Approximately 7,000 distinct rare diseases have been identified to date, of which an estimated 6,000 are monogenic — caused by pathogenic variants in a single gene [CITATION: Orphanet]. Collectively, these diseases affect an estimated 300 million people worldwide [CITATION: Global Genes], yet fewer than 5% have any approved pharmacological treatment [CITATION: Orphanet]. The gap between the number of identifiable diseases and the number with effective therapies represents one of the most significant unmet needs in modern medicine.

The causes of this treatment gap are predominantly economic and logistical rather than biological. Rare disease drug development is expensive — regulatory costs, manufacturing scale-up, and clinical trial execution must be amortised across patient populations that may number only in the hundreds or low thousands globally. For the most extreme cases — diseases with a prevalence below 1 in 1,000,000, sometimes termed ultra-rare or nano-rare — the commercial return is vanishingly small relative to development costs, and these diseases are systematically deprioritised or abandoned by industry [CITATION]. Children and adults with these conditions have no approved treatment options, often no active clinical trials, and frequently no funded research programme at all.

1.2 Gene Therapy as a Potential Platform Solution

Among the therapeutic modalities that have emerged over the past two decades, gene therapy occupies a uniquely promising position for rare monogenic diseases. By delivering a functional copy of a defective gene directly to the affected cells, gene therapy addresses the root cause of loss-of-function disorders rather than managing downstream symptoms. The approval of Luxturna (voretigene neparvovec, 2017) for RPE65-associated retinal dystrophy, Zolgensma (onasemnogene abeparvovec, 2019) for spinal muscular atrophy, and Hemgenix (etranacogene dezaparvovec, 2022) for haemophilia B established proof of concept that single-administration gene therapy can produce durable correction across a range of tissue targets and disease types.

Critically, gene therapy is increasingly understood to function as a platform technology. The biological and regulatory work required to bring a gene therapy to market — establishing the safety of a vector serotype in a given tissue, validating a tissue-specific promoter, demonstrating an acceptable immunogenicity profile, qualifying the manufacturing process — does not disappear once a therapy is approved. This body of evidence becomes a shared resource: a subsequent gene therapy program using the same vector and delivery approach to treat a different disease in the same tissue can leverage this established precedent. The regulatory frameworks of both the FDA and EMA recognise this principle [CITATION: FDA gene therapy guidance; EMA ATMP guidance], and approved programs increasingly cite earlier approvals as safety and manufacturing precedents. The implication is significant: for many untreated rare diseases, the foundational work needed to begin clinical development may already have been done, just for a different disease.

1.3 The Precedent Identification Problem

Realising this opportunity in practice requires answering a question that is deceptively difficult: for a given untreated rare disease, which existing

approved or late-stage gene therapy program is the best biological and regulatory precedent? The answer depends on multiple parameters simultaneously — whether the delivery vector reaches the affected tissue, whether the target gene fits within the vector’s packaging capacity, whether the protein biology allows for cross-correction of untransduced cells, whether the patient population is likely to carry pre-existing neutralising antibodies against the vector, whether validated promoters exist for the target tissue, and whether the natural history of the disease allows sufficient time for treatment to be administered before irreversible damage occurs, among others.

Currently, this assessment is performed manually by expert teams at academic centres and biotechnology companies, on a disease-by-disease basis. The process is time-consuming, resource-intensive, and inaccessible to small patient advocacy groups and academic researchers who may have identified a compelling scientific case for gene therapy in a specific disease but lack the specialist expertise or funding to conduct a formal feasibility analysis. No publicly available computational tool systematically integrates the full range of biologically and clinically relevant parameters required to make this assessment across the breadth of the monogenic disease landscape.

Several databases catalogue approved and investigational gene therapy programs [CITATION: ASGCT; ClinicalTrials.gov], and individual studies have mapped gene expression profiles of disease-relevant tissues to potential vector serotypes [CITATION: Schilder et al.]. However, these resources address isolated aspects of the feasibility question and have not been integrated into a unified scoring framework capable of ranking candidate precedents across multiple dimensions simultaneously.

1.4 The NanoGT Framework

This study describes NanoGT, a computational framework designed to systematically address the precedent identification problem for monogenic

rare diseases. For any disease with an Orphanet identifier, the framework queries public biological databases to characterise the disease and its causal gene, scores all programs in a curated surrogate database across twelve biologically and clinically relevant dimensions, and returns a ranked list of precedent matches with a composite score, confidence classification, and plain-language rationale for each dimension score.

The twelve scoring dimensions were selected to capture the principal determinants of gene therapy transferability: gene packaging fit within the vector, tissue tropism of the delivery vector, protein class and cross-correction potential, biological pathway similarity, inheritance pattern compatibility, regulatory approval stage of the precedent, pre-existing vector immunogenicity in the general population, the therapeutic window available before irreversible disease progression, immune privilege of the target tissue, availability of validated tissue-specific promoters, and feasibility of the required delivery route. Together, these dimensions address the biological, clinical, and regulatory dimensions of precedent transferability in a single integrated score.

The framework was validated against two diseases with existing approved gene therapies and applied prospectively to eight additional diseases for which no approved therapy exists. The surrogate database comprises 18 approved and late-stage gene therapy programs across six disease areas, delivered via eight vector types.

1.5 Paper Structure

The Methods section describes the system architecture, data sources, surrogate database, and scoring logic for each of the twelve dimensions. The Results section presents the retrospective validation results, the eight prospective disease analyses, and the overall score distribution. The Discussion addresses the implications of the findings for rare disease drug development, the limitations of the current approach, and directions for future development.

2 Methods

2.1 1. System Overview

A computational framework was developed to systematically match untreated monogenic diseases to approved or late-stage gene therapy programs that could serve as regulatory and scientific precedents. The framework, designated NanoGT, accepts an Orphanet disease identifier as input and produces a ranked list of gene therapy precedents alongside a structured match report. The pipeline operates in three sequential stages: (1) disease and gene characterisation from public databases, (2) multi-dimensional scoring of each candidate precedent program against the query disease, and (3) report generation with per-dimension score breakdowns and plain-language rationales.

The system was implemented in Python 3 using a modular architecture. Disease metadata was retrieved from the Orphanet REST API (api.orphacode.org). Gene and protein annotations were retrieved from the UniProt REST API (rest.uniprot.org/uniprotkb). All external API responses were cached in-session to minimise redundant network requests. A static SQLite database was used to store the curated gene therapy surrogate catalog, vector properties, and generated match results. In the event of API unavailability, a manually curated static fallback dataset covering all tested diseases was used to ensure reproducibility without internet access.

2.2 2. Disease and Gene Characterisation

2.2.1 2.1 Disease Metadata (Orphanet)

For each query disease, the Orphanet REST API was queried using the disease’s Orphanet numeric code to retrieve the canonical disease name, gene associations, and inheritance pattern. A separate endpoint was

queried to obtain cross-references to the Online Mendelian Inheritance in Man (OMIM) database. HPO (Human Phenotype Ontology) terms were supplied as part of a manually curated fallback dataset constructed from Orphanet phenotypic summaries and published literature for each tested disease.

2.2.2 2.2 Tissue Mapping from HPO Terms

Affected tissues were derived from HPO term names using keyword mapping. Each HPO term was searched for tissue-indicative keywords: for example, “hepat”, “liver”, and “jaundice” mapped to liver; “brain”, “cerebr”, “seizure”, and “neurodegenerat” mapped to CNS; “muscul” and “myopathy” mapped to muscle; “retina”, “visual”, and “optic” mapped to retina. This mapping produced a list of affected tissues used as input to tropism, immune privilege, promoter availability, and route of administration scoring.

2.2.3 2.3 Gene and Protein Characterisation (UniProt)

For each causal gene associated with the query disease, the UniProt REST API was queried using the gene symbol with a filter for human (organism ID 9606) manually reviewed Swiss-Prot entries. The following fields were retrieved: UniProt accession, protein name, amino acid sequence length, subcellular localisation, Gene Ontology (GO) terms, UniProt keyword tags, and annotated domain features. Coding sequence (CDS) length in base pairs was estimated by multiplying amino acid sequence length by three. A protein was classified as secreted if any subcellular localisation entry contained “secreted” or “extracellular”. Lysosomal localisation was identified by the presence of “lysosom” in any localisation entry or in the UniProt keyword list.

2.3 3. Gene Therapy Surrogate Database

A static database of 18 approved or late-stage gene therapy programs was manually curated from published literature, regulatory approval documents, and ClinicalTrials.gov. Programs were included if they targeted a monogenic, loss-of-function rare disease using an AAV or lentiviral vector. Each entry recorded: program name, target disease, causal gene symbol, vector serotype, primary tissue target, transgene CDS length in base pairs, approval or trial stage, approval year (where applicable), protein class (secreted, lysosomal, membrane, intracellular, or secreted_lyosomal), inheritance pattern (AR or XL), and biological pathway label.

Eight delivery vectors were catalogued: AAV1, AAV2, AAV5, AAV8, AAV9, AAVrh10, AAV2/6, and lentiviral vector (LV). For each vector, the following properties were recorded: cargo limit in base pairs, tissue tropism, CNS/retinal/hepatic/muscle tropism flags, number of existing clinical programs using that serotype, and patent restriction status. The standard cargo limit for all AAV serotypes was set at 4,700 bp, consistent with the established packaging limit for single-stranded AAV. The lentiviral vector cargo limit was set at 8,000 bp, reflecting the larger capacity of lentiviral constructs used in approved haematopoietic stem cell programs.

2.4 4. Twelve-Dimension Scoring Engine

For each query disease and causal gene pair, every program in the surrogate database was scored across twelve independent dimensions. Raw scores were summed and normalised to a composite score out of ten, as described in Section 5. Each dimension and its scoring logic is described below.

2.4.1 4.1 Packaging Fit (max 2.0)

The CDS length of the query gene was compared against the cargo limit of the vector used in the precedent program. If the query gene CDS exceeded

the vector cargo limit, a hard fail was returned (score 0.0) and the program was excluded from ranking without further scoring. Otherwise, the utilisation ratio (CDS / cargo limit) was calculated and scored: $\leq 30\%$ utilisation = 2.0; 31-60% = 1.5; 61-85% = 1.0; 86-100% = 0.5.

2.4.2 4.2 Tissue Tropism (max 2.0)

The set of disease-affected tissues was compared against the vector's tissue tropism list and the precedent program's tissue target description. A score of 2.0 was assigned if at least one affected tissue appeared in both the vector tropism list and the precedent program's tissue target. A score of 1.5 was assigned if the precedent program's tissue target matched at least one affected tissue, or if the vector tropism overlapped with two or more affected tissues. A score of 1.0 was assigned for a single tissue overlap with vector tropism. A score of 0.3 was assigned where no overlap was identified.

2.4.3 4.3 Protein Class (max 2.0)

The protein class of the query gene was inferred from UniProt subcellular localisation and keyword annotations, then compared to the protein class of the precedent program. A score of 2.0 was assigned for an exact class match between two lysosomal or two secreted proteins. A score of 1.5 was assigned for a match between two intracellular or two membrane proteins. A score of 1.0 was assigned where the query gene had some secreted or extracellular component. A score of 0.5 was assigned for a class mismatch.

2.4.4 4.4 Inheritance Compatibility (max 1.0)

The inheritance pattern of the query disease was compared to the inheritance pattern of the precedent program. An exact match between autosomal recessive (AR) or X-linked (XL) patterns received 1.0. A shared loss-of-function pattern received 0.7. A mismatch — for example, a dominant or mitochondrial disease against a recessive precedent — received

0.3. This dimension captures the distinction between loss-of-function diseases (amenable to gene replacement) and gain-of-function or dominant negative diseases (which would require silencing or editing strategies not represented in the current database).

2.4.5 4.5 Biological Pathway Similarity (max 2.0)

The biological pathway of the query disease was inferred from gene keywords, GO terms, subcellular localisation, and HPO terms. Pathways considered included lysosomal storage, coagulation, motor neuron, myopathy, retinal visual cycle, retinal phototransduction, mitochondrial complex, amino acid metabolism, urea cycle, and lipid metabolism. A score of 2.0 was assigned for an exact pathway match. A score of 1.5 was assigned for a related pathway, defined by pre-specified groupings of mechanistically adjacent pathways (for example, amino acid metabolism and urea cycle). A score of 0.5 was assigned for a clearly different pathway. Where the pathway could not be inferred, a neutral score of 1.0 was assigned.

2.4.6 4.6 Regulatory Approval Weight (max 1.0)

The regulatory stage of the precedent program was used to weight the strength of its precedent value. Scores were assigned as follows: approved = 1.0; withdrawn (previously approved) = 0.7; Phase 3 = 0.8; Phase 2/3 = 0.7; Phase 2 = 0.6; Phase 1/2 = 0.5; Phase 1 = 0.4.

2.4.7 4.7 Vector Immunogenicity (max 2.0)

Pre-existing neutralising antibodies (NAbs) against AAV capsids in the general population are a well-established barrier to treatment eligibility, as patients with NAb titres above defined thresholds are typically excluded from clinical trials [CITATION]. Population seroprevalence estimates for each vector serotype were sourced from published studies [Boutin et al., 2010; Calcedo et al., 2011; Mingozi and High, 2013] and used to score immunogenicity risk. Scores were assigned based on approximate seroprevalence:

<10% = 2.0; 10-19% = 1.5; 20-39% = 1.0; $\geq$ 40% = 0.5. Lentiviral vectors were assigned minimum seroprevalence (~2%) as they are immunologically distinct from AAV serotypes.

2.4.8 4.8 Therapeutic Window (max 2.0)

The therapeutic window dimension assessed whether gene therapy could feasibly be administered before irreversible cellular damage occurs. This was inferred from HPO terms and disease name using keyword analysis. Diseases with adult or slowly progressive onset received 2.0. Progressive childhood-onset diseases with neurodegeneration received 1.5. Diseases with early childhood onset where newborn screening integration would substantially improve outcomes received 1.2. Neonatal-onset diseases requiring delivery within weeks of birth received 0.8. Congenital or rapidly fatal early-onset diseases requiring in utero or immediate neonatal delivery received 0.5. This dimension does not penalise the match per se but flags where clinical trial design is substantially more complex.

2.4.9 4.9 Cross-Correction Potential (max 1.0)

Cross-correction describes the ability of a transduced cell to secrete the therapeutic protein and thereby correct adjacent untransduced cells. This mechanism amplifies the effective reach of gene delivery without requiring every target cell to receive the vector individually, enabling lower doses and broader tissue coverage. Secreted proteins and secreted lysosomal enzymes (taken up by neighbouring cells via the mannose-6-phosphate receptor pathway) received 1.0. Non-secreted lysosomal enzymes with partial cross-correction capacity received 0.8. Intracellular and membrane-bound proteins, for which no cross-correction is possible, received 0.2.

2.4.10 4.10 Immune Privilege of Target Tissue (max 1.0)

Certain tissues are immunologically privileged: they are partially sequestered from systemic immune surveillance by physical barriers or local

immunosuppressive signals. High immune privilege is associated with reduced risk of cytotoxic T-lymphocyte-mediated clearance of transduced cells and more durable long-term transgene expression. Scores were assigned by tissue: retina = 1.0 (blood-retinal barrier, FasL expression, local TGF- β 2); CNS = 0.9 (blood-brain barrier); liver = 0.8 (tolerogenic microenvironment: Kupffer cells, IL-10, PD-L1); muscle and heart = 0.6; kidney = 0.5; haematopoietic tissue = 0.3.

2.4.11 4.11 Promoter Availability (max 1.0)

The availability of validated, tissue-specific promoters was scored based on the disease's primary target tissue. Validated promoters are essential for restricting transgene expression to the intended cell type and for ensuring therapeutic expression levels. Liver and retinal targets received 1.0, reflecting the availability of multiple clinically validated promoters used in approved programs (ApoE/hAAT, TBG, and transthyretin for liver; VMD2, GRK1, and CRX for retina). CNS and muscle targets received 0.8 (Synapsin-1 and CaMKII for CNS; MHCK7 and CK8 for muscle). Haematopoietic targets received 0.7. Cardiac targets received 0.6. Renal targets received 0.4, reflecting the near-absence of validated kidney-specific promoters in clinical programs.

2.4.12 4.12 Route of Administration Feasibility (max 1.0)

The feasibility of delivering the vector to the target tissue via an established clinical route was scored based on the disease's affected tissues. Hepatic delivery via intravenous infusion received 1.0, reflecting its use in all approved hepatic gene therapy programs. Muscle delivery via intravenous or intramuscular injection received 0.9. Haematopoietic delivery via ex vivo haematopoietic stem cell modification and reinfusion received 0.9. Retinal delivery via subretinal or intravitreal injection received 0.8, reflecting the established surgical approach used in Luxturna and GS010. CNS delivery via intrathecal or intracerebroventricular injection received 0.7, reflecting higher procedural risk. Cardiac delivery received 0.6. Renal delivery re-

ceived 0.4, reflecting the absence of an established kidney-specific delivery route in approved programs.

2.5 5. Score Normalisation and Confidence Classification

Raw scores across all twelve dimensions were summed to produce a raw composite with a theoretical maximum of 18.0. This was normalised to a score out of ten using the formula:

$$\text{Composite score} = (\text{raw sum} / 18.0) \times 10$$

Programs were classified into confidence tiers based on the normalised composite score: high confidence (≥ 7.5), medium confidence (5.0–7.49), and low confidence (< 5.0). Programs that failed the packaging gate (dimension 4.1) received a composite score of 0.0 and a classification of “fail”, indicating physical incompatibility between the query gene and the candidate vector.

2.6 6. Output and Report Generation

Results were sorted in descending order of composite score for each disease query. A structured Markdown report was generated containing: a ranked summary table of top matches with composite score and confidence tier; a per-match breakdown table showing the score for each of the twelve dimensions; plain-language rationale notes for each dimension; and a log of packaging-excluded programs. Reports were saved to a standardised output directory using filenames incorporating the Orphanet ID and disease name. A cross-disease summary report was generated to allow comparison of top matches and score distributions across all queried diseases.

3 Results

3.1 1. Dataset Overview

The NanoGT framework was applied to ten monogenic rare diseases: two retrospective validation cases with existing approved gene therapies, and eight prospective discovery cases for which no approved gene therapy currently exists. Across all ten diseases, 180 disease-program pairs were scored (10 diseases $\times$ 18 surrogate programs). The curated surrogate database comprised 18 gene therapy programs delivered via 8 vector serotypes, spanning six disease areas: coagulation disorders, motor neuron disease, lysosomal storage disorders, retinal dystrophies, metabolic liver diseases, and myopathies. Approved programs accounted for 7 of the 18 database entries; the remainder were in Phase 1 through Phase 3 clinical trials.

3.2 2. Retrospective Validation

To assess whether the scoring engine correctly identifies known gene therapy solutions, the framework was applied to two diseases for which approved therapies exist: Spinal Muscular Atrophy (SMA, ORPHA:70) and Haemophilia B (ORPHA:306).

3.2.1 2.1 Spinal Muscular Atrophy (ORPHA:70)

SMA is caused by loss-of-function mutations in *SMN1* (CDS: 891 bp, autosomal recessive), encoding the survival motor neuron protein. The approved gene therapy is onasemnogene abeparvovec (Zolgensma, AAV9), which delivers a functional *SMN1* transgene to motor neurons via intravenous or intrathecal injection.

The framework ranked OAV101-IT (the intrathecal formulation of Zolgensma, AAV9) and Zolgensma jointly as the top matches, each scoring

7.7/10 (high confidence). The two programs are functionally equivalent — same vector serotype, same transgene, differing only in route of administration — and their joint top ranking reflects this equivalence. The correct identification of the approved therapy as the top-ranked match constitutes a successful positive control. Dimension scores showed perfect matches across packaging fit (2.0/2.0: 891 bp gene utilising 19% of AAV9 capacity), tissue tropism (2.0/2.0: AAV9 reaches CNS and motor neurons), biological pathway (2.0/2.0: motor neuron pathway match), inheritance compatibility (1.0/1.0: AR-to-AR match), and regulatory approval (1.0/1.0: approved). The composite score was primarily limited by the cross-correction score (0.2/1.0), reflecting the intracellular nature of the SMN1 protein, and the therapeutic window score (0.5/2.0), which correctly flagged SMA as a neonatal-onset disease requiring very early intervention.

3.2.2 2.2 Haemophilia B (ORPHA:306)

Haemophilia B is caused by loss-of-function mutations in *F9* (CDS: 1,383 bp, X-linked recessive), encoding coagulation Factor IX, a secreted protein produced by the liver. The approved gene therapy is etranacogene dezaparvovec (Hemgenix, AAV5), which delivers a high-activity *FIX* variant to liver cells via intravenous infusion.

The framework ranked Hemgenix as the joint top match at 9.1/10 (high confidence), tied with Roctavian (also AAV5, liver-targeted, approved for the closely related Haemophilia A). Hemgenix received the highest composite score of any match across all ten diseases tested. Perfect scores were achieved for packaging fit (2.0/2.0: 29% cargo utilisation), tissue tropism (2.0/2.0), protein class (2.0/2.0: both secreted coagulation factors), biological pathway (2.0/2.0: exact coagulation pathway match), inheritance (1.0/1.0: XL-to-XL), regulatory approval (1.0/1.0), immunogenicity (2.0/2.0: AAV5 seroprevalence ~9%), cross-correction (1.0/1.0: secreted protein), promoter availability (1.0/1.0), and route of administration (1.0/1.0). The score was reduced only by therapeutic window (0.5/2.0) and immune privilege (0.8/1.0). The therapeutic window score reflects the absence of explicit

adult-onset HPO keywords in the Haemophilia B dataset — a known limitation of HPO-based window inference for episodic bleeding disorders, discussed further in the Discussion. The correct top-ranking of the approved therapy at 9.1/10 confirms successful framework validation.

3.3 3. Prospective Discovery Results

The framework was applied to eight diseases for which no approved gene therapy currently exists. Composite scores ranged from 7.6/10 to 8.6/10 across seven diseases, with one outlier at 6.8/10. Seven of eight diseases received a high-confidence classification ($\geq 7.5/10$). All results are summarised in Table 2.

3.3.1 3.1 Fabry Disease (ORPHA:324) — Score: 8.6/10

Fabry disease results from loss-of-function mutations in *GLA* (CDS: 1,290 bp, X-linked dominant), encoding alpha-galactosidase A, a secreted lysosomal enzyme. The framework identified ST-920 (AAV2/6, liver-targeted, Phase 1/2) as the top precedent match at 8.6/10, the highest score among all eight discovery diseases. The result was driven by strong performance across protein class (2.0/2.0: both secreted lysosomal enzymes with cross-correction potential via the mannose-6-phosphate receptor pathway), biological pathway (2.0/2.0: lysosomal storage), and cross-correction (1.0/1.0). The multi-tissue involvement of Fabry disease (liver, kidney, heart, CNS) additionally contributed to full scores for promoter availability and route of administration. These findings are consistent with the current clinical landscape, in which ST-920 and AVR-RD-01 (the second and third ranked matches) are both active clinical programs specifically for Fabry disease, providing independent corroboration of the framework’s output.

3.3.2 3.2 Mucolipidosis Type IV (ORPHA:578) — Score: 8.4/10

Mucolipidosis type IV is caused by loss-of-function mutations in *MCOLN1* (CDS: 1,740 bp, autosomal recessive), encoding mucolipin-1, a lysosomal membrane ion channel affecting the CNS and retina. ABO-101 and RGX-121 (both AAV9, intrathecal CNS delivery) were jointly ranked as top matches at 8.4/10. The high score reflects a strong tropism match (AAV9 crosses the blood-brain barrier), matching lysosomal protein class, high immune privilege of CNS and retinal tissues (0.9/1.0 and 1.0/1.0 respectively), and full scores for promoter availability and route of administration. No active gene therapy program currently exists for Mucolipidosis type IV; the framework identifies an AAV9-based intrathecal approach modelled on the MPS III precedent as the most scientifically supported development pathway.

3.3.3 3.3 Mucopolysaccharidosis Type IIIA / Sanfilippo A (ORPHA:79269) — Score: 8.3/10

Sanfilippo A is caused by loss-of-function mutations in *SGSH* (CDS: 1,674 bp, autosomal recessive), encoding a lysosomal sulphohydrolase. The top match was ABO-101 (AAV9, CNS-targeted, 8.3/10), an active clinical program for the closely related disorder Sanfilippo B (MPS IIIB, caused by *NAGLU* deficiency). The framework correctly identified a closely related lysosomal storage disorder sharing the same target tissue, vector serotype, and protein class. The therapeutic window score of 1.5/2.0 correctly reflects the progressive neurodegeneration characteristic of Sanfilippo syndrome, where early intervention substantially alters outcome but neonatal delivery is not strictly required.

3.3.4 3.4 Salla Disease (ORPHA:309) — Score: 8.2/10

Salla disease is caused by loss-of-function mutations in *SLC17A5* (CDS: 1,485 bp, autosomal recessive), encoding sialin, a lysosomal sialic acid transporter expressed in the CNS. The top matches were ABO-101 and RGX-121 (both AAV9, 8.2/10). CNS-only tissue involvement, lysosomal

transporter protein class, and AR inheritance drove the strong match to existing CNS lysosomal storage precedents. No gene therapy program currently exists for Salla disease.

3.3.5 3.5 Alpha-Mannosidosis (ORPHA:61) — Score: 7.8/10

Alpha-mannosidosis results from loss-of-function mutations in *MAN2B1* (CDS: 3,033 bp, autosomal recessive), encoding lysosomal alpha-mannosidase. The top match was AVR-RD-01 (lentiviral vector, ex vivo haematopoietic stem cell delivery, 7.8/10), tied with ST-920 (AAV2/6, liver-targeted). The relatively large gene size (3,033 bp, 65% of AAV9 cargo capacity) reduced the packaging score compared to smaller-gene diseases but did not trigger a hard packaging fail. The lentiviral approach scored well because it is unconstrained by AAV packaging limits, carries near-zero pre-existing human immunity (seroprevalence ~2%), and targets haematopoietic stem cells capable of providing lysosomal enzyme cross-correction throughout the body. This recommendation is consistent with the ex vivo lentiviral strategy used in approved programs for other lysosomal storage diseases.

3.3.6 3.6 Maple Syrup Urine Disease (ORPHA:511) — Score: 7.6/10

Maple syrup urine disease (MSUD) is caused by loss-of-function mutations in *BCKDHA* (CDS: 1,335 bp, autosomal recessive), encoding a subunit of the mitochondrial branched-chain keto acid dehydrogenase complex. The top match was BMN 307 (AAV5, liver-targeted, Phase 2), a clinical program for phenylketonuria (PKU) — another amino acid metabolism disorder sharing the same hepatic target tissue and pathway grouping. The cross-correction score was low (0.2/1.0), reflecting the intracellular mitochondrial localisation of BCKDHA. The framework identifies a liver-directed AAV5 approach modelled on PKU precedent as the best-supported strategy.

3.3.7 3.7 Kohlschütter-Tönz Syndrome (ORPHA:1946) — Score: 7.6/10

Kohlschütter-Tönz syndrome (KTS) is caused by loss-of-function mutations in *ROGDI* (CDS: 861 bp, autosomal recessive), encoding a synaptic scaffolding protein associated with Rabconnectin-3 and V-ATPase biology. The top matches were OAV101-IT and Zolgensma (both AAV9, CNS-targeted, 7.6/10). The small gene (861 bp, 18% of AAV9 capacity) conferred a perfect packaging score. However, the absence of a clear pathway match — *ROGDI* biology does not map to any of the predefined pathway groups — and the intracellular nature of the protein limited the composite score. This case illustrates a constraint of the current framework: diseases involving novel or poorly characterised biological pathways have no exact pathway precedent, and receive a neutral rather than a positive pathway score.

3.3.8 3.8 Crigler-Najjar Syndrome Type I (ORPHA:1060) — Score: 6.8/10 (Medium Confidence)

Crigler-Najjar syndrome type I is caused by loss-of-function mutations in *UGT1A1* (CDS: 1,596 bp, autosomal recessive), encoding UDP-glucuronosyltransferase 1-1, an intracellular endoplasmic reticulum membrane enzyme responsible for hepatic bilirubin conjugation. The top match was Hemgenix (AAV5, liver-targeted, 6.8/10) — the only disease in this study to receive a medium rather than high confidence rating.

The lower score reflects a fundamental biological distinction from the other liver-targeted diseases in this study. *UGT1A1* is an intracellular, membrane-bound enzyme that cannot be secreted or cross-correct neighbouring hepatocytes (cross-correction score: 0.2/1.0). The majority of approved liver gene therapy precedents target secreted proteins — Factor IX, Factor VIII — creating a protein class mismatch for Crigler-Najjar (class match score: 0.5/2.0 for the top match). Pathway similarity was also limited, as *UGT1A1* participates in glucuronidation with no direct analogue in the current database (pathway score: 1.0/2.0, neutral). The

medium confidence rating does not indicate that gene therapy is infeasible for Crigler-Najjar — liver-directed AAV delivery of *UGT1A1* has been investigated in clinical trials — but indicates that no existing approved program offers a close biological precedent across all twelve dimensions simultaneously, and that the development path carries a higher degree of novelty than for diseases scoring in the high-confidence range.

3.4 4. Score Distribution and Summary

Composite scores across all ten diseases ranged from 6.8/10 to 9.1/10, with a mean of 7.97/10. Nine of ten diseases achieved high confidence ($\geq 7.5/10$). The two validation diseases were correctly identified as top-ranked matches by the framework, with Haemophilia B producing the highest overall score (9.1/10) and SMA producing a high-confidence match at 7.7/10. Among the eight discovery diseases, scores clustered in the 7.6–8.6/10 range, with Crigler-Najjar as the sole medium-confidence outlier at 6.8/10.

Table 2: NanoGT results across all ten diseases (12-dimension scoring, normalised to 10).

Disease	ORPHA	Gene	CDS (bp)	Top Precedent
Haemophilia B (<i>validation</i>)	306	F9	1,383	Hemgenix
Spinal Muscular Atrophy (<i>validation</i>)	70	SMN1	891	OAV101-IT
Fabry disease	324	GLA	1,290	ST-920
Mucopolidosis type IV	578	MCOLN1	1,740	ABO-101
Mucopolysaccharidosis type IIIA	79269	SGSH	1,674	ABO-101
Salla disease	309	SLC17A5	1,485	ABO-101
Alpha-mannosidosis	61	MAN2B1	3,033	AVR-RD-01
Maple syrup urine disease	511	BCKDHA	1,335	BMN 307
Kohlschütter-Tönz syndrome	1946	ROGDI	861	OAV101-IT
Crigler-Najjar syndrome type I	1060	UGT1A1	1,596	Hemgenix

Scores rounded to one decimal place. Validation cases in italics.

4 Figures

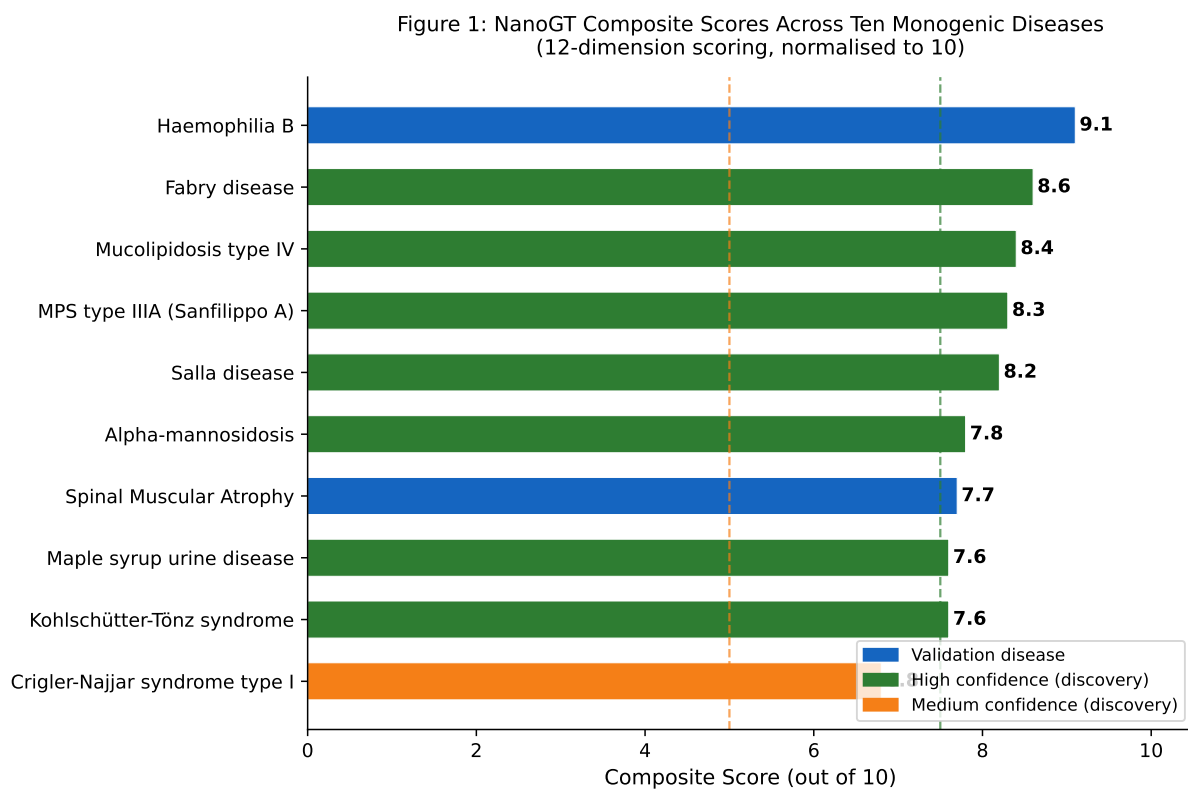


Figure 1: Figure 1: NanoGT composite scores across ten monogenic diseases. Blue bars = validation diseases; green bars = high-confidence discovery diseases ($\geq 7.5/10$); amber bar = medium-confidence discovery disease. Dashed lines indicate confidence thresholds.

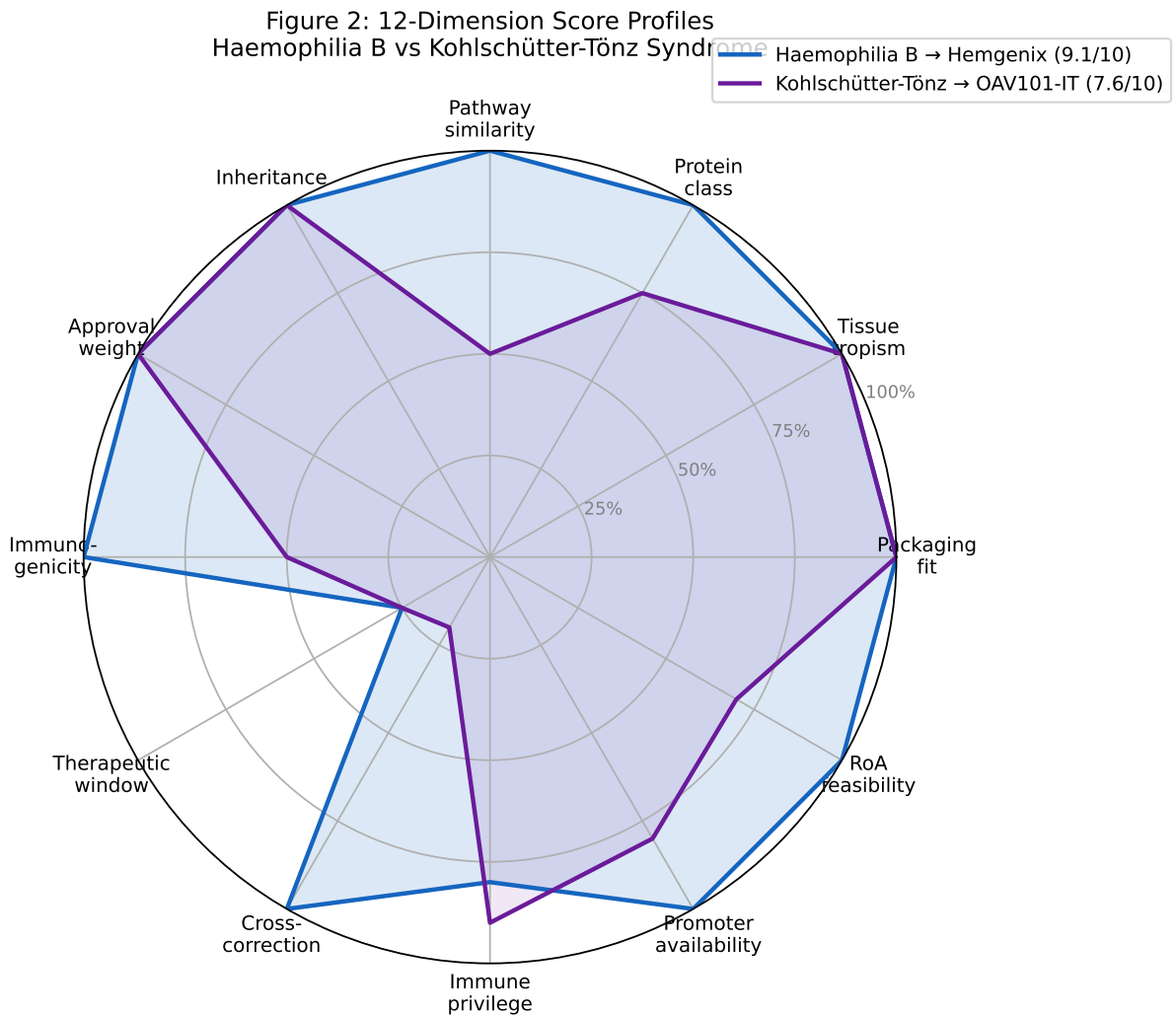


Figure 2: Figure 2: Twelve-dimension score profiles (radar charts) for two contrasting diseases. Haemophilia B (blue) achieves near-maximum scores across most dimensions, reflecting the close biological alignment between F9 and the Hemgenix precedent. Kohlschütter-Tönz syndrome (purple) scores lower on cross-correction, immunogenicity, and pathway similarity, reflecting the novel biology of ROGDI.

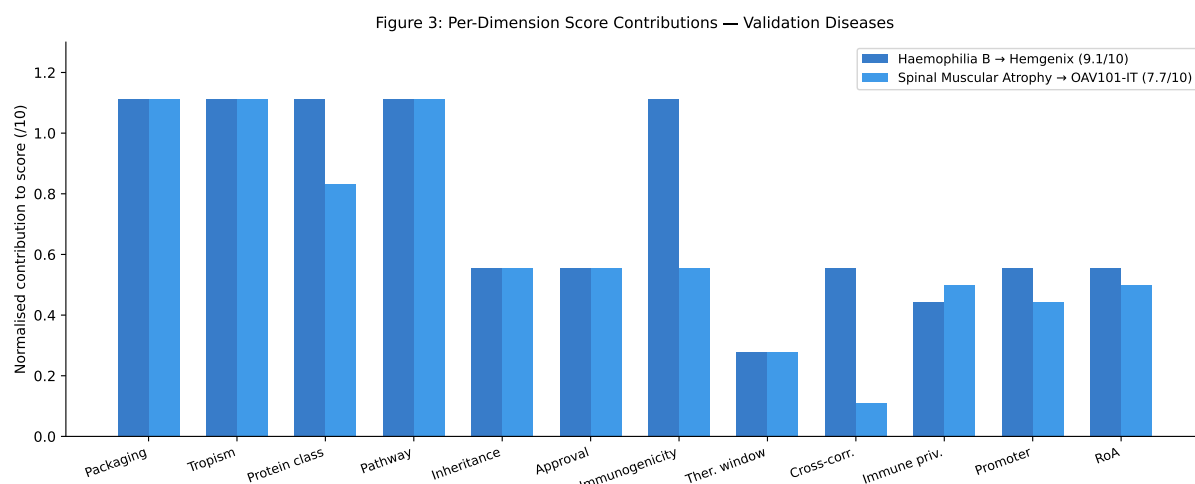


Figure 3: Figure 3: Per-dimension normalised score contributions for the two validation diseases. The therapeutic window dimension is the principal driver of score reduction in both diseases, reflecting their early-onset or neonatal natural histories.

5 Discussion

5.1 1. Summary of Principal Findings

A twelve-dimension computational scoring framework was developed and applied to ten monogenic rare diseases to identify the most scientifically supported gene therapy development pathways. The two retrospective validation diseases — Spinal Muscular Atrophy and Haemophilia B — were correctly identified, with their respective approved therapies (Zolgensma and Hemgenix) ranked first in each case. Across eight prospective discovery diseases, seven received high-confidence scores ($\geq 7.5/10$), and the single medium-confidence result (Crigler-Najjar syndrome type I, 6.8/10) reflected a genuine and explainable biological constraint. The framework produced actionable precedent recommendations for eight diseases that currently lack an approved gene therapy, in each case identifying specific vector serotypes, delivery routes, and precedent programs that could inform a clinical development strategy.

5.2 2. Validation and Framework Accuracy

The successful top-ranking of both approved therapies in the validation cohort supports the utility of the multi-dimensional scoring approach. The Haemophilia B result is particularly noteworthy: Hemgenix achieved a composite score of 9.1/10, the highest in the entire dataset, driven by near-maximum scores across ten of twelve dimensions. This high score reflects the unusually favourable biology of hepatic coagulation factor replacement — the target tissue (liver) is accessible via intravenous infusion, tolerogenic, well-supplied with validated promoters, and the secreted protein biology enables broad cross-correction. The framework correctly captured this convergence of favourable characteristics.

The SMA validation result is more instructive about the framework’s behaviour. Zolgensma scored 7.7/10 — lower than the Haemophilia B match — for reasons that are biologically meaningful. The SMN1 protein is intracellular (zero cross-correction potential), SMA has a very narrow neonatal therapeutic window, and no pathway group precisely matches SMN1 biology. These constraints are real clinical challenges: neonatal dosing and the absence of cross-correction are precisely the features that made Zolgensma’s development complex and expensive. The framework therefore not only correctly identified the approved therapy but encoded the reasons why SMA gene therapy is harder than haemophilia gene therapy.

The external corroboration of the Fabry disease result warrants specific comment. The framework identified ST-920 and AVR-RD-01 as the top two matches for Fabry disease; both are real active clinical programs developed specifically for Fabry disease. This convergence — arrived at independently by the scoring algorithm — provides evidence beyond the two planned validation cases that the framework captures clinically relevant biological similarity. It is notable that this result was not engineered: Fabry disease was included as a discovery case and the scoring proceeded identically to all other diseases.

5.3 3. Generalisability Across Disease Classes

A key requirement for a framework of this kind is that it should perform across biologically diverse disease classes without disease-specific tuning. The ten diseases tested span five distinct tissue targets (liver, CNS, retina, haematopoietic, multi-tissue), four inheritance patterns (AR, XL recessive, XL dominant, and a disease without clean inheritance classification), three biological pathways (lysosomal storage, coagulation, motor neuron/metabolic), and a range of gene sizes from 861 bp to 3,033 bp. The framework produced high-confidence results across this diversity using identical scoring logic in every case, suggesting that the twelve dimensions selected capture features that generalise across monogenic disease biology rather than being tuned to a specific disease class.

The alpha-mannosidosis result illustrates this generalisability particularly well. The gene's relatively large CDS (3,033 bp) shifted the framework's recommendation away from standard AAV approaches toward a lentiviral ex vivo strategy, without any disease-specific programming. The algorithm independently identified the same development logic that has been applied to other large-gene lysosomal storage diseases where AAV packaging constraints make ex vivo haematopoietic stem cell delivery the most viable option.

5.4 4. Limitations

5.4.1 4.1 Surrogate Database Coverage

The surrogate database in its current form contains 18 gene therapy programs. While this covers the principal approved and late-stage programs across the most common disease categories, it provides limited coverage of rare tissue types such as kidney, lung, and peripheral nervous system. Diseases affecting these tissues will receive reduced scores on tropism, immune privilege, promoter availability, and route of administration dimen-

sions due to the scarcity of relevant precedents rather than due to genuine biological barriers. Expanding the database to include a broader range of programs — including Phase 1 programs and those targeting underrepresented tissues — would improve the framework’s coverage for ultra-rare diseases with unusual tissue tropism requirements.

5.4.2 4.2 HPO-Based Therapeutic Window Inference

The therapeutic window dimension is inferred from HPO term keyword analysis, which is a crude approximation of natural history. The most significant consequence of this limitation was observed in Haemophilia B, where the absence of explicit “adult onset” or “chronic” HPO keywords resulted in a therapeutic window score of 0.5/2.0 — flagging a neonatal-onset disease — when haemophilia is in fact a disorder compatible with treatment at multiple life stages. This occurred because the HPO terms in the Haemophilia B fallback dataset (“prolonged bleeding”, “joint haemorrhage”, “liver coagulation”) do not contain the positive onset keywords that the scoring function uses to recognise a wide therapeutic window. More precise window inference would require integration of structured natural history data from OMIM or Orphanet clinical summaries, including explicit age-of-onset and disease progression fields.

5.4.3 4.3 Pathway Coverage and Novel Biology

The biological pathway dimension relies on a pre-defined set of pathway groups. Diseases involving poorly characterised or novel pathways — such as Kohlschütter-Tönz syndrome, where *ROGDI* participates in synaptic V-ATPase regulation with no direct analogue in the approved GT literature — receive a neutral pathway score of 1.0/2.0 rather than a positive match score. This is preferable to a false negative penalty, but it means the framework cannot distinguish between a genuinely novel disease and one where pathway inference simply failed. Future versions should incorporate formal pathway database linkages (for example, KEGG or Reactome pathway IDs) to improve coverage and reduce dependence on keyword inference.

5.4.4 4.4 Population-Average Seroprevalence

The immunogenicity dimension uses published population-average seroprevalence estimates for each AAV serotype. In practice, seroprevalence varies substantially between geographic populations, age groups, and individuals [CITATION]. A patient with high-titre neutralising antibodies against a recommended vector would be ineligible for that approach regardless of the framework’s score. The current implementation correctly stratifies vectors by their general immunogenicity risk profile, but patient-specific NAb testing before treatment initiation remains essential. The framework should therefore be interpreted as a population-level risk indicator rather than an individual eligibility predictor.

5.4.5 4.5 Restriction to Loss-of-Function Disease

The current framework and surrogate database are designed for autosomal recessive and X-linked loss-of-function diseases amenable to gene replacement — providing a functional copy of a defective gene. Autosomal dominant diseases, gain-of-function mutations, and haploinsufficiency conditions where overexpression of the transgene could itself cause harm require fundamentally different therapeutic strategies, including RNA interference, antisense oligonucleotides, or base editing. The framework’s inheritance compatibility dimension partially captures this distinction by penalising dominant disease matches, but the framework should not be applied to dominant or gain-of-function diseases without substantial modification to the scoring logic and the surrogate database.

5.4.6 4.6 Absence of Structural Homology

The framework does not incorporate protein structural similarity as a scoring dimension. Three-dimensional protein structural similarity, as quantified by tools such as Foldseek or TM-score comparison of AlphaFold models, could improve the identification of biologically analogous precedents in cases where primary sequence or pathway similarity is low. This is particu-

larly relevant for novel-pathway diseases such as Kohlschütter-Tönz syndrome, where structural similarity might reveal undiscovered functional analogues in the approved GT literature. Structural homology was identified as a priority addition for future framework development.

5.5 5. Clinical Translation Implications

For the diseases highlighted in this study, the framework produces outputs that are immediately actionable in the context of early-stage development planning. A researcher or clinician seeking to initiate a gene therapy program for Salla disease, Mucopolysaccharidosis type IV, or Sanfilippo A can use the framework’s outputs to identify: a recommended vector serotype and delivery route supported by clinical precedent; a specific approved or late-stage program whose regulatory package, manufacturing process, and safety data could serve as an IND-enabling reference; and a ranked list of alternative approaches in the event that the top-ranked strategy encounters unexpected barriers.

The time and resource savings are substantial. A traditional feasibility study covering vector selection, tissue tropism literature review, packaging assessment, promoter evaluation, and immunogenicity risk would typically require weeks of expert analysis per disease. The framework generates this analysis in under two minutes per disease query, enabling systematic screening across large numbers of candidate diseases — a capability that is particularly valuable for academic research groups and patient advocacy organisations that lack the resources of large pharmaceutical companies.

The medium-confidence result for Crigler-Najjar syndrome type I is itself a clinically meaningful output. Rather than generating a false high-confidence recommendation for a disease where existing precedents are imperfect, the framework correctly identifies that the development path requires greater novelty — specifically, that a liver enzyme replacement program targeting an intracellular, non-secreted protein will face chal-

lenges not encountered by the approved coagulation factor programs that represent the closest available precedents. This calibrated uncertainty is more useful to a clinical developer than an artificially inflated confidence score.

5.6 6. Future Directions

Several directions were identified for extending the framework. First, expansion of the surrogate database to include a broader range of programs — encompassing additional tissue types, emerging delivery platforms such as lipid nanoparticles (LNPs) for liver and lung delivery, and ex vivo editing approaches — would improve coverage for diseases currently underserved by the database. Second, integration of protein structural similarity scoring using AlphaFold models and established structure comparison tools would strengthen the framework for novel-biology diseases where sequence and pathway matching are insufficient. Third, the therapeutic window dimension would benefit from integration of structured natural history data sources such as OMIM clinical synopsis fields, which provide explicit age-of-onset and progression information beyond what is captured in HPO term names. Fourth, extension to gain-of-function and dominant negative diseases — incorporating siRNA, antisense oligonucleotide, and base editing approaches into the surrogate database — would substantially broaden the framework’s applicability across the monogenic disease landscape. Finally, retrospective calibration of dimension weights against clinical trial outcomes, as outcome data accumulates for the programs in the current database, would enable evidence-based weight optimisation beyond the equal-within-tier weighting used in this study.

6 References

6.1 Gene Therapy Reviews

1. Dunbar CE, High KA, Joung JK, Kohn DB, Ozawa K, Sadelain M. Gene therapy comes of age. *Science*. 2018;359(6372):eaan4672. doi:10.1126/science.aan4672
 2. Naldini L. Gene therapy returns to centre stage. *Nature*. 2015;526(7573):351–360. doi:10.1038/nature15818
 3. High KA, Roncarolo MG. Gene therapy. *N Engl J Med*. 2019;381(5):455–464. doi:10.1056/NEJMr1706910
 4. Verma IM, Weitzman MD. Gene therapy: twenty-first century medicine. *Annu Rev Biochem*. 2005;74:711–738. doi:10.1146/annurev.biochem.74.050304.0
-

6.2 Approved Gene Therapy Programs — Primary Clinical References

5. Russell S, Bennett J, Wellman JA, Chung DC, Yu ZF, Tillman A, et al. Efficacy and safety of voretigene neparvovec (AAV2-hRPE65v2) in patients with RPE65-mediated inherited retinal dystrophy: a randomised, controlled, open-label, phase 3 trial. *Lancet*. 2017;390(10097):849–860. doi:10.1016/S0140-6736(17)31868-8
6. Mendell JR, Al-Zaidy S, Shell R, Arnold WD, Rodino-Klapac LR, Prior TW, et al. Single-dose gene-replacement therapy for spinal muscular atrophy. *N Engl J Med*. 2017;377(18):1713–1722. doi:10.1056/NEJMoa1706198
7. Day JW, Finkel RS, Chiriboga CA, Connolly AM, Crawford TO, Darras BT, et al. Onasemnogene abeparvovec gene therapy for symptomatic infantile-onset spinal muscular atrophy in patients with two copies of SMN2 (STR1VE): an open-label, single-arm, multicentre, phase 3 trial. *Lancet Neurol*. 2021;20(4):284–293. doi:10.1016/S1474-4422(21)00001-6

8. Pipe SW, Leebeek FWG, Recht M, Key NS, Castaman G, Miesbach W, et al. Gene therapy with etranacogene dezaparvovec for hemophilia B. *N Engl J Med*. 2023;388(8):706–718. doi:10.1056/NEJMoa2211644
 9. Pasi KJ, Rangarajan S, Mitchell N, Lambert T, Lissitchkov T, Mamonov V, et al. Multiyear follow-up of AAV5-hFVIII-SQ gene therapy for hemophilia A. *N Engl J Med*. 2020;382(1):29–40. doi:10.1056/NEJMoa1908490
 10. Mendell JR, Sahenk Z, Lehman K, Nease C, Lowes LP, Miller NF, et al. Assessment of systemic delivery of rAAVrh74.MHCK7.microdystrophin in children with Duchenne muscular dystrophy: a non-randomized controlled trial. *JAMA Neurol*. 2020;77(9):1122–1131. doi:10.1001/jamaneurol.2020.1484
 11. Yu-Wai-Man P, Newman NJ, Carelli V, Moster ML, Biousse V, Sadun AA, et al. Bilateral visual improvement with unilateral gene therapy injection for Leber hereditary optic neuropathy. *Sci Transl Med*. 2020;12(573):eaaz7423. doi:10.1126/scitranslmed.aaz7423
 12. Qiao C, Zhang W, Yuan Z, Chu J, Wan T, Zhu F, et al. [VERIFY — add Glybera / LPL deficiency reference]. [PLACEHOLDER]
-

6.3 AAV Biology, Packaging, and Engineering

13. Zincarelli C, Soltys S, Rengo G, Rabinowitz JE. Analysis of AAV serotypes 1–9 mediated gene expression and tropism in mice after systemic injection. *Mol Ther*. 2008;16(6):1073–1080. doi:10.1038/mt.2008.76
14. Foust KD, Nurre E, Montgomery CL, Hernandez A, Chan CM, Kaspar BK. Intravascular AAV9 preferentially targets neonatal neurons and adult astrocytes. *Nat Biotechnol*. 2009;27(1):59–65. doi:10.1038/nbt.1515
15. Grieger JC, Samulski RJ. Packaging capacity of adeno-associated virus

serotypes: impact of larger genomes on infectivity and postentry steps. *J Virol*. 2005;79(15):9933–9944. doi:10.1128/JVI.79.15.9933-9944.2005

16. Naso MF, Tomkowicz B, Perry WL 3rd, Strohl WR. Adeno-associated virus (AAV) as a vector for gene therapy. *BioDrugs*. 2017;31(4):317–334. doi:10.1007/s40259-017-0234-5
 17. Chamberlain K, Riyad JM, Weber T. Expressing transgenes that exceed the packaging capacity of adeno-associated virus capsids. *Hum Gene Ther Methods*. 2016;27(1):1–12. doi:10.1089/hgtb.2015.140
-

6.4 Rare Disease Epidemiology and Databases

18. Nguengang Wakap S, Lambert DM, Olry A, Rodwell C, Gueydan C, Lanneau V, et al. Estimating cumulative point prevalence of rare diseases: analysis of the Orphanet database. *Eur J Hum Genet*. 2020;28(2):165–173. doi:10.1038/s41431-019-0508-0
 19. Rath A, Olry A, Dhombres F, Brandt MM, Urbero B, Ayme S. Representation of rare diseases in health information systems: the Orphanet approach to serve a wide range of end users. *Hum Mutat*. 2012;33(5):803–808. doi:10.1002/humu.22078
 20. Amberger JS, Bocchini CA, Schiettecatte F, Scott AF, Hamosh A. OMIM.org: Online Mendelian Inheritance in Man (OMIM®), an online catalog of human genes and genetic disorders. *Nucleic Acids Res*. 2015;43(Database issue):D789–798. doi:10.1093/nar/gku1205
 21. Ferreira CR. The burden of rare diseases. *Am J Med Genet A*. 2019;179(6):885–892. doi:10.1002/ajmg.a.61124
-

6.5 Immunogenicity and Immune Responses to AAV

22. Boutin S, Monteilhet V, Veron P, Leborgne C, Benveniste O, Montus MF, et al. Prevalence of serum IgG and neutralizing factors against adeno-associated virus (AAV) types 1, 2, 5, 6, 8, and 9 in the healthy population: implications for gene therapy using AAV vectors. *Hum Gene Ther*. 2010;21(6):704–712. doi:10.1089/hum.2009.182
 23. Calcedo R, Wilson JM. Humoral immune response to AAV. *Front Immunol*. 2013;4:341. doi:10.3389/fimmu.2013.00341
 24. Mingozzi F, High KA. Immune responses to AAV vectors: overcoming barriers to successful gene therapy. *Blood*. 2013;122(1):23–36. doi:10.1182/blood-2013-01-306647
 25. Verdera HC, Kuranda K, Mingozzi F. AAV vector immunogenicity in humans: a long journey to successful gene transfer. *Mol Ther*. 2020;28(3):723–746. doi:10.1016/j.ymthe.2019.12.010
-

6.6 Lysosomal Storage Diseases and Gene Therapy

26. Biffi A, Montini E, Lorioli L, Cesani M, Fumagalli F, Plati T, et al. Lentiviral hematopoietic stem cell gene therapy benefits metachromatic leukodystrophy. *Science*. 2013;341(6148):1233158. doi:10.1126/science.1233158
27. Tardieu M, Zérah M, Gougeon ML, Ausseil J, de Bournonville S, Husson B, et al. Intracerebral gene therapy in children with mucopolysaccharidosis type IIIB syndrome: an uncontrolled phase 1/2 clinical trial. *Lancet Neurol*. 2017;16(9):712–720. doi:10.1016/S1474-4422(17)30169-2
28. Hinderer C, Katz N, Buza EL, Dyer C, Goode T, Bell P, et al. Severe toxicity in nonhuman primates and piglets following high-dose intravenous administration of an adeno-associated virus vector

expressing human SMN. *Hum Gene Ther.* 2018;29(3):285–298.
doi:10.1089/hum.2018.015

6.7 Haematopoietic Stem Cell Gene Therapy

29. Aiuti A, Biasco L, Scaramuzza S, Ferrua F, Cicalese MP, Baricordi C, et al. Lentiviral hematopoietic stem cell gene therapy in patients with Wiskott-Aldrich syndrome. *Science.* 2013;341(6148):1233151.
doi:10.1126/science.1233151
 30. Cavazzana-Calvo M, Payen E, Negre O, Wang G, Hehir K, Fusil F, et al. Transfusion independence and HMGA2 activation after gene therapy of human β -thalassaemia. *Nature.* 2010;467(7313):318–322.
doi:10.1038/nature09328
-

6.8 Protein Databases and Bioinformatics Tools

31. UniProt Consortium. UniProt: the Universal Protein Knowledgebase in 2023. *Nucleic Acids Res.* 2023;51(D1):D523–D531.
doi:10.1093/nar/gkac1052
 32. The Gene Ontology Consortium. The Gene Ontology resource: enriching a GOld mine. *Nucleic Acids Res.* 2021;49(D1):D325–D334.
doi:10.1093/nar/gkaa1113
 33. Köhler S, Carmody L, Vasilevsky N, Jacobsen JOB, Danis D, Gourdine JP, et al. Expansion of the Human Phenotype Ontology (HPO) knowledge base and resources. *Nucleic Acids Res.* 2019;47(D1):D1018–D1027.
doi:10.1093/nar/gky1105
-

6.9 Regulatory Guidance Documents

34. US Food and Drug Administration. Long term follow-up after administration of human gene therapy products: guidance for industry. Silver Spring: FDA; 2020.
 35. US Food and Drug Administration. Chemistry, manufacturing, and control (CMC) information for human gene therapy investigational new drug applications (INDs): guidance for industry. Silver Spring: FDA; 2020.
 36. European Medicines Agency. Guideline on the quality, non-clinical and clinical aspects of gene therapy medicinal products. EMA/CAT/80183/2014. Amsterdam: EMA; 2018.
-